

Individual Report - Design Thinking

Introduction

The company chosen for this report is Apple, and the product being focused on is the original iPod. Apple, with the iPod and its adjacent service, iTunes, revolutionized how we listen to music on the go. The iPod effectively killed the Sony Walkman, the de facto on-the-go music player of the time. Without the iPod, Apple may never have become the company that went on to create the iPhone.

Company Background

Apple Inc. is one of the most recognizable brands in the world and is currently the most valuable company on the planet. However, Apple comes from humble beginnings. It is one of the earliest examples of a company that achieved worldwide reach after being started in a garage and bootstrapped by its founders.

Apple was founded by Steve Jobs and Steve Wozniak in 1976 with a small investment from Ronald Wayne. Ronald left the company that year and sold his 10% stake. Apple Inc. was incorporated in January 1997, with Mike Markkula providing seed funding of \$250,000.

The Apple II was the company's first product after the Apple I and was the first widely distributed microcomputer in the world, selling over five to six million units over the next decade (Levy & Montevirgen, 1998). Even at this early stage in the company's history, the focus on design and innovation was evident. Steve Jobs was famously picky about even small details, such as the consistency in soldering and the alignment of wires in the circuit.

Following the Apple II, Apple released a slew of products that were commercial busts, such as the Lisa and Macintosh, which eventually led to the ousting of its infamous founder, Steve Jobs, from the company. Jobs would go on to found NeXT. After Jobs' ousting, Apple faced challenges in both its legal department and its ability to innovate. Furthermore, competition from Microsoft and IBM made it increasingly difficult for Apple to compete. Apple released the Newton in 1993; although a flop, it was the first device to integrate handwriting recognition nearly 30 years before handwriting recognition finally reached a point where it was both usable and reliable. The Apple Newton could be considered a product ahead of its time—a product released to a market that wasn't ready and with technology that was not advanced enough.

In 1996, Apple purchased NeXT Software Inc., bringing back its infamous co-founder, Steve Jobs, in an advisory role until he was recruited to finally lead the company he co-founded in mid-1997. Thus began Apple's renaissance as it became the undisputed leader in consumer electronics.

Markets Served

Operating in over 176 countries, Apple is currently valued at \$3.63 trillion and operates in the consumer electronics space, with various retail and digital services supporting its hardware and software business. Apple's top moneymaker is the iPhone, with its digital services coming in second.

Value Proposition of the product

Starting in 1998, a slew of digital MP3 players hit the market. Steve Jobs saw this as an opportunity and believed that Apple could make a much better product.

The story of the iPod starts with the launch of iTunes in January 2001, which enabled the writing and management of CDs on one's computer. Eventually, iTunes expanded into a digital music retail store.

Paired with iTunes, the iPod was a significant improvement over competing products. Through an exclusive deal with Toshiba for a disc drive, the iPod was able to store about 1,000 songs—1,000 songs in your pocket. With a compact form factor and, most importantly, the enablement of nibble search and control through the click wheel, there was nothing like the iPod on the market. Although Apple and Steve Jobs had believed in the power of creating closed systems since Apple's inception in 1976, the combination of iTunes and the iPod can be seen as the beginning of Apple's famous ecosystem, where hardware and software work perfectly together and a slew of services exclusive to their devices enhance their capabilities and streamline the customer experience.

What is the need that is being addressed?

The iPod was not the first digital music player to hit the market. In 2001, the record business was facing severe competition from digital file-sharing services like Napster. The industry decided that rather than innovating and trying to compete, it would pursue litigation instead (Forde, 2021). Thus, there was a need to find a middle ground where record companies would still get paid and users would have the ability to store music digitally.

The iPod paired with iTunes allowed the music industry to find a middle ground with the digital or technology industry, where the record companies would still be paid for the music they owned but through iTunes rather than CDs.

How was the product conceived using design thinking

Apple's former Vice President of Industrial Design, Jony Ive, said, "So much of what we do is worry about the smallest of details." Unlike products from other companies that stand out for the plethora of features they contain, Apple takes a completely different approach; its products are noted for the features they lack. This is not due to a lack of technical capabilities; rather, it relies on following the principle of "less is more." In this approach, every element or feature on the product must have a purpose and earn its place.

Paul Mercer, whose company implemented the UI software for Apple, said, "The iPod is very simple-minded... it really doesn't do much other than let you navigate your music." This leads back to the concept of job-to-be-done (JTBD); in this case, the job was simply to play music on the go.

The creation of the iPod can be traced back to understanding the customer and creating a device that allowed people to listen to music digitally on the go. At the time, MP3 players were bulky, had limited storage, and featured a complicated file-transfer process. Apple recognized this and set out to create a device that could alleviate all these problems with the technology available.

The leadership articulated three very simple goals for the iPod: "For the hardware: A thousand songs in your pocket; for the software: so easy that your Mother could do it; and for the project: on the shelves in eight months!" (Fischer, 2021). These goals were not easy to accomplish but helped clearly define the basis of the product they set out to create and provided a clear vision to the iPod team.

For the iPod team, Jobs selected many individuals who are now notable, including Jony Ive and his design team, Jon Rubenstein, and Tony Fadell (now credited as the inventor of the iPod). Thus, a team capable of achieving the iPod's goals was created.

Like most of Apple's products, the prototyping process for the iPod is shrouded in mystery. However, Apple is known to follow the "10 to 3 to 1" prototyping method. Under Jony Ive, the industrial design team employed a rapid ideation process in which they first brainstormed ten divergent ideas. Out of these ten, three were evaluated based on their technical feasibility and user appeal; finally, the chosen design underwent relentless iteration (Elmansy, 2014).

At the time of the original iPod's creation, MP3 players tended to look like TV remotes; the iPod stood out for its clean and minimalist design, with its most notable feature being the clickable scroll wheel. Great emphasis was placed on the design of the scroll wheel. It featured a bearing in the center that allowed the wheel to spin smoothly and prevented it from rocking side to side. Due to the short timeline available to pull the design together, the early prototypes left significant room for optimization. Steve Jobs even dunked an iPod prototype in an aquarium and pointed to the air bubbles rising from the device to show that the device could become even smaller (Month, 2022). This instance again highlights Jobs' and Apple's insistence on design and their desire for constant innovation.

How was the organization in the company structured to support an innovation culture?

At the time of the iPod's creation, Apple was organized in a manner completely different from other companies. Under Steve Jobs, there were almost no silos separating teams. To this day, Apple employees rave about the culture of deep collaboration and concurrent engineering. This means that, unlike in most organizations where products are passed from team to team during different stages of the product development process, all development stages at Apple were simultaneous and organic rather than sequential. All departments—from the design, hardware, and software teams to the marketing team—worked on the product at once (Fischer, 2021). Jony Ive said, “By keeping the core team small and investing significantly in [prototyping] tools and process[es], we can work with a level of collaboration that seems particularly rare. In fact, the memory of how we work will endure beyond the products of our work” (Fischer, 2021).

Where did they view this product: core, adjacency, or transformational?

The iPod was viewed as transformational because of its ability to revolutionize the music industry and change the way consumers viewed MP3 players. iTunes, however, was considered an adjacent product. Although transformational in the way the service allowed users to burn CDs onto their computers and share music, it was not a major driver of revenue for the company, despite iTunes owning 70% of the digital music retail space. The user experience enabled by the combination of iTunes and the iPod was critical to the iPod's commercial success.

How was their commercialization plan constructed?

Steve Jobs understood that the potential market for the iPod was limited by the number of songs available on the device. Jobs reportedly castigated the executives of record companies, telling them that they needed an easy-to-access music store. This led the CEO of Warner Music to suggest that each track could sell for 99 cents, with Apple pocketing 22 cents for every track sold on iTunes. Eventually, both Sony and Universal bought into the deal as well, enabling iPod users to choose from a nearly unlimited selection of tracks for their devices. Initially, however, the record labels insisted that the iPod should remain a Mac-only device, given that Apple at the time only had 5% of global computer sales. The success of the iPod eventually allowed Apple to persuade the record labels to permit making the iPod compatible with Windows.

Even during Jobs' first tenure, Apple was known for its marketing—a reputation that holds true to this day. Apple spent a considerable amount of money marketing the iPod, even diverting funds from the Mac to bolster the iPod campaign. Apple Stores, which marked Apple's debut in the retail space, were launched only five months prior to the iPod's release. These stores depended on foot traffic and eventually became a major driver for Apple's growth and its emergence as a household name. Apple Stores proved significant in making the iPod a commercial success.

Business plan for profitability?

Aside from Apple's integrated ecosystem (hardware, software, services) and the creation of retail Apple Stores, one major driver of profitability then—and to this day—has been the operational excellence of Tim Cook. Tim Cook joined Apple in 1998 with the assignment to “clean up the atrocious state of Apple's manufacturing, distribution, and supply apparatus,” which allowed inventory that once took months to clear to drop to only a few days. This same efficiency was applied to the manufacturing, sales, and distribution of the iPod, enabling Apple to significantly drive down per-unit cost and increase profitability.

Your personal reflection if this was successful or not.

What was done well? Why?

A lot went well in the development of the iPod. It allowed Apple to become the behemoth it is today. The iPod was an excellent marriage of design, hardware, and software. Furthermore, the eight-month development timeline for such a monumental

product is quite astonishing. The iPod also proved to be a significant investment in the future, as its development directly guided the creation of the iPhone.

What didn't work well? Why?

The iPod was not an overnight success; it sold only 25,000 units in the first year. It took time and effort to iterate the iPod into a product worthy of mass adoption. Furthermore, it required significant investment by Apple in the form of marketing. Apple was also late to the market, which meant that they were playing catch-up, and at launch, the iPod didn't seem revolutionary enough for mass adoption. Additionally, the early restrictions on Windows compatibility severely hindered early sales of the iPod, although this issue was eventually resolved in Apple's favor.

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